(a) 如果损失函数的Hessian矩阵是半正定的，

即损失函数是凸函数，只要找到一个极小值点，$J(\theta)=-\frac{1}{m}\sum[y^i \ln h_\theta(x^i)+(1-y^i)\ln(1-h_\theta(x^i))]$

就一定是全局最优解。

$$\nabla J(\theta)=\frac{\partial J}{\partial h}\cdot\frac{\partial h}{\partial z}\cdot\frac{\partial z}{\partial \theta}$$

$$=\left(\frac{y^{(i)}}{h_\theta(x^{(i)})}-\frac{1-y^{(i)}}{1-h_\theta(x^i)}\right)\cdot h_\theta(x^i)(1-h_\theta(x^i))\cdot x_i$$

$$=\frac{1}{m}\sum_{i=1}^{m}(h_\theta(x_i)-y_i)x_i$$

$$H=\nabla J(\theta)=\frac{1}{m}\sum_{i=1}^{m}[\nabla^2 h_\theta(x_i)](x_i)^T$$

$$=\frac{1}{m}\sum_{i=1}^{m}(h_\theta(x_i)(1-h_\theta(x_i)))x_i\cdot(x_i)^T$$

$$z^THz=\frac{1}{m}\sum_{i=1}^{m}\underbrace{(h_\theta(x_i)(1-h_\theta(x_i)))}_{A}\underbrace{(z^Tx)(x^Tz)}_{B}$$

由于 $h_\theta\in(0,1)$ 则 $A>0$

B平方项 ≥ 0

即 $z^THz\geq 0$ 恒成立，即H半正定

(c)

$$P(y=1|x)=\frac{P(x|y=1)P(y=1)}{P(x|y=0)P(y=0)+P(x|y=1)P(y=1)}$$

分子分母同除 $P(x|y=1)P(y=1)$

$$P(y=1|x)=\frac{1}{1+\exp\left(-\ln\left(\frac{P(x|y=1)P(y=1)}{P(x|y=0)P(y=0)}\right)\right)}$$

$$\ln\left(\frac{P(x|y=1)P(y=1)}{P(x|y=0)P(y=0)}\right)=\ln\frac{P(x|y=1)}{P(x|y=0)}+\ln\frac{P(y=1)}{P(y=0)}$$

$$=\underbrace{(\mu_1-\mu_0)^T\Sigma^{-1}}_{\theta^T}x+\underbrace{\frac{1}{2}\mu_0^T\Sigma^{-1}\mu_0-\frac{1}{2}\mu_1^T\Sigma^{-1}\mu_1+\ln\frac{\varphi}{1-\varphi}}_{\theta_0}$$

$$P(y=1|x)=\frac{1}{1+\exp(-(\theta^Tx+\theta_0))}$$

在GDA中，两类别共享一个协方差矩阵时，其后验概率具有logistic function形式